

***Brock Biology of Microorganisms, 15e (Madigan et al.)***  
**Chapter 31 Vectorborne and Soilborne Bacterial and Viral Diseases**

31.1 Multiple Choice Questions

1) Which rickettsial disease is sometimes transmitted by food, such as milk, from infected domestic animals?

- A) ehrlichiosis
- B) typhus
- C) Q fever
- D) Rocky Mountain spotted fever

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.3

2) A student travels to Brazil by boat on the Amazon River, without visiting the doctor prior to traveling for recommended vaccinations for travel to South America. The student becomes ill during the trip and is quarantined. Which of the diseases listed below does the student most likely have?

- A) hantavirus
- B) yellow fever
- C) Ebola
- D) Q fever

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 31.5

3) Eradication of certain diseases is nearly impossible, because we cannot

- A) control or immunize all vector animals.
- B) control or immunize all reservoir animals.
- C) eradicate organisms completely from the soil.
- D) control or immunize all vector and reservoir animals or eradicate organisms completely from the soil.

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 31.9

4) Which of the following flaviviruses can be prevented by vaccination?

- A) yellow fever only
- B) dengue fever only
- C) Zika only
- D) All of these answer choices flaviviruses can be prevented by vaccination.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.5

5) Animal diseases transmissible to humans are known as

- A) vectors.
- B) zoonoses.
- C) ferals.
- D) radicals.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.1

6) If you were exposed to a sick animal that was found to have negri bodies in its nervous system in a postmortem examination, how would you be treated to prevent disease?

- A) passive and active immunization against rabies
- B) antibiotics to treat Lyme disease
- C) antiviral medication
- D) observation to detect tick or lice bites

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 31.1

7) Which insect functions in the host-to-host transmission of typhus?

- A) rat flea
- B) body or head louse
- C) dog tick
- D) deer tick

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.3

8) Which of the following would be the best strategy to prevent the spread of yellow fever and dengue fever into the southern United States?

- A) mosquito avoidance and eradication
- B) vaccination
- C) monitoring bird populations
- D) tick avoidance

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 31.5

9) What type of diseases require biosafety level (BSL) 4 procedures and facilities when working with samples that may be infected?

- A) highly infectious diseases
- B) diseases with high mortality
- C) diseases for which no treatment or vaccine exist
- D) highly infectious diseases with high mortality for which no treatment or vaccine exist

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.2

10) Which of the following diseases CANNOT easily be controlled by vector elimination?

- A) yellow fever
- B) dengue fever
- C) rabies
- D) plague

Answer: C

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 31.5

11) Very few preventative measures are taken against anthrax, because

- A) the general population is not exposed to infected materials or animals regularly.
- B) non-weaponized anthrax causes only a mild flu-like illness.
- C) no vaccines are effective.
- D) the causative agent *Bacillus anthracis* mutates rapidly.

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 31.8

12) Animal-transmitted diseases are generally spread to accidental hosts by

- A) direct contact.
- B) aerosols.
- C) bites.
- D) direct contact, aerosols, and bites.

Answer: D

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 31.1

13) Soldiers or refugees in crowded unsanitary conditions are most susceptible to which rickettsial disease?

- A) Lyme disease
- B) typhus
- C) Q fever
- D) Rocky Mountain spotted fever

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 31.3

14) The hantavirus contains a \_\_\_\_\_ genome.

- A) ssRNA
- B) dsRNA
- C) ssDNA
- D) dsDNA

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.2

15) Hantaviruses are most commonly transmitted by \_\_\_\_\_ of virus-contaminated rodent \_\_\_\_\_.

- A) inhalation / saliva
- B) inhalation / excreta
- C) ingestion / saliva
- D) ingestion / excreta

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.2

16) Of the rickettsias, *C. burnetti* is unusual because it

- A) can be cultured in pure culture outside of a host.
- B) is transmitted by a vector.
- C) can be cultured in laboratory animals.
- D) does not grow intracellularly.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 30.3

17) The rickettsias are divided into groups based loosely on

- A) the types of clinical disease they produce.
- B) their insect vectors.
- C) taxonomy.
- D) their biochemical requirements.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.3

18) Which of the following pathogens is spread by infected lice?

- A) *Borrelia burgdorferi*
- B) *Plasmodium falciparum*
- C) *Rickettsia prowazekii*
- D) *Yersinia pestis*

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.3

19) Which rickettsial disease can cause illness that progresses to severe symptoms including fever, headache, and endocarditis?

- A) Q fever
- B) typhus
- C) Rocky Mountain spotted fever
- D) scrub typhus

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.3

20) Which disease(s) has/have been mistakenly diagnosed as measles or scarlet fever due to similarities in the corresponding rash caused by those diseases?

- A) Lyme disease
- B) Q fever
- C) rickettsial diseases
- D) typhus

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.3

21) What virulence factor(s) of *Yersinia pestis* prevent(s) phagocytosis by cells of the immune system?

- A) the capsule
- B) leukocidins
- C) murine toxin
- D) V and W antigens

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.7

22) Many West Nile virus infections are not properly identified because

- A) the disease mimics malaria.
- B) the symptoms are often subacute.
- C) the disease is frequently fatal.
- D) most physicians and diagnosticians are not aware of this disease.

Answer: B

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 31.6

23) Which of the following diseases causes an influenza-like illness that can progress to include pneumonia?

- A) malaria
- B) Q fever
- C) typhus
- D) Lyme disease

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.3

24) Rocky Mountain Spotted Fever can be

- A) prevented using vaccination.
- B) spread by lice.
- C) diagnosed by culturing in the laboratory without a host animal.
- D) treated effectively with tetracycline or chloramphenicol.

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.3

25) The disease whose symptoms are most closely related to the symptoms of chronic syphilis is

- A) Rocky Mountain spotted fever.
- B) chronic Lyme disease.
- C) typhus.
- D) HPS.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.4

26) Tetanus develops from a puncture wound because

- A) the organism's endospores germinate in anoxic conditions.
- B) the organism gains access to the circulatory system and then invades the nervous system.
- C) the organism digests deep connective tissues.
- D) the organism needs access to the circulatory system and deep connective tissues.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.9

27) Murine toxin is a \_\_\_\_\_ that is a respiratory inhibitor and lethal to mice.

- A) *Yersinia pestis* endotoxin
- B) *Rickettsia prowazekii* endotoxin
- C) *Yersinia pestis* exotoxin
- D) *Rickettsia prowazekii* exotoxin

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.7

28) In bubonic plague, buboes are swellings formed in the lymph nodes and filled with

- A) *Yersinia pestis*.
- B) *Rickettsia prowazekii*.
- C) *Xenopsylla cheopis*.
- D) *Clostridium tetani*.

Answer: A

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.7

29) The primary disease reservoir for plague is

- A) fleas.
- B) humans.
- C) rats.
- D) mosquitoes.

Answer: C

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.7

30) A person is most likely to contract diseases such as Rocky Mountain spotted fever and Ehrlichiosis if he or she visits/participates in

- A) camping and hiking in the backcountry.
- B) metropolitan cities.
- C) snow skiing.
- D) European countries.

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 31.3

31) The most severe and fatal form of anthrax is

- A) cutaneous anthrax.
- B) inhalation anthrax.
- C) intestinal anthrax.
- D) zoonotic anthrax.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.8

32) Which of the following was an emerging mosquito-borne disease in the early 2000s and has now spread to every state in the United States?

- A) malaria
- B) Rocky Mountain spotted fever
- C) typhus
- D) West Nile fever

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.6

33) The causative agent of Lyme disease is \_\_\_\_\_, which is carried primarily by the \_\_\_\_\_.

- A) a flavivirus / *Ixodes* sp. (deer tick)
- B) *Yersinia pestis* / *Xenopsylla cheopis* (rat flea)
- C) *Rickettsia prowazekii* / *Culex* species (mosquito)
- D) *Borrelia burgdorferi* / *Ixodes* sp. (deer tick)

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.4

34) Hantavirus pulmonary syndrome is a viral respiratory disease caused by contact with

- A) rat fleas.
- B) rodents.
- C) mosquitoes.
- D) bats.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.2

35) The causative agent of dengue fever is

- A) *Rickettsia prowazekii*.
- B) a flavivirus.
- C) *Borrelia burgdorferi*.
- D) *Clostridium perfringens*.

Answer: B

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.5

36) Controlling most rickettsial diseases involves managing

- A) the vector.
- B) human behavior.
- C) immunization strategies.
- D) food safety procedures.

Answer: A

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 31.3

37) Hantaviruses are related to viruses such as \_\_\_\_\_ and \_\_\_\_\_.

- A) human immunodeficiency virus (HIV) / Hepatitis C
- B) Hepatitis A / Hepatitis B
- C) human immunodeficiency virus (HIV) / Influenza A
- D) Lassa fever virus / Ebola virus

Answer: D

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.6

### 31.2 True/False Questions

1) Emerging diseases are a problem worldwide, even in highly developed regions of the world.

Answer: TRUE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 31.5

2) If Lyme disease is not treated early, it can infect the central nervous system.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.4

3) In different geographic locations, different tick and *Borrelia* spp. may be involved in the transmission and cause of Lyme disease.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.4



4) The majority of cases of human plague in the United States occur in the northeastern states.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.7

5) Anthrax is an emerging disease for which there is no treatment or vaccine available.

Answer: FALSE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 31.8

6) Inhalation anthrax is the only form of anthrax for which there is no effective treatment.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.8

7) Both yellow and dengue fever can be prevented by vaccines.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.5

8) Certain *Clostridium* spp. can be found in the human gut, but *Clostridium tetani* is almost exclusively present in soil.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.9

9) Deer flies (*Chrysops* spp.) are a major vector in the spread of typhus.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.3

10) Travelers should be vaccinated against yellow fever when traveling to Sub-Saharan Africa or South America.

Answer: TRUE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 31.5

11) Only high-risk individuals such as scientists working with the organism, livestock workers, and military personnel are vaccinated for anthrax.

Answer: TRUE

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 31.8

12) Rabies infection in humans leads to symptoms a few days after exposure.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.1

13) Epizootic diseases often occur in wild animals on a periodic or cyclic basis.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.1

14) Human granulocytic anaplasmosis is associated with typhus.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.3

15) The hantaviruses are occasionally spread by person-to-person transmission, although exposure to infected rodent excreta is the most common route of infection.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.2

16) The hantavirus diagnosis can be made from several immunological techniques.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.2

17) There are several different virus-specific treatments and vaccines available to treat or prevent hantavirus infections.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.2

18) *Rickettsia prowazekii* is transmitted to humans when a louse bite becomes contaminated with louse feces.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.3

19) Tetracycline and chloramphenicol are the drugs of choice for the treatment of many of the rickettsial diseases.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.3

20) Lyme disease is found across the United States, but is most common in the northeast.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.4

21) A PCR assay has been developed for the detection of the bacterium (*Borrelia burgdorferi*) that causes Lyme disease.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.4

22) Tetanus is highly contagious through person-to-person contact.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.9

23) Wild animals are a natural reservoir for *Clostridium tetani*.

Answer: FALSE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.9

24) Virtually all cases of tetanus occur in individuals who have failed to receive sufficient tetanus toxoid booster immunizations.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.9

25) Septicemic plague usually causes death before a diagnosis can be made.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.7

26) The study of hantaviruses requires biosafety level (BSL) 4 precautions.

Answer: TRUE

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.2

### 31.3 Essay Questions

1) Explain why a potentially rabid dog is held for up to two weeks to check for clinical signs of rabies. Are humans more similar or dissimilar in this regard?

Answer: Dogs have a short incubation time when infected by rabies virus. Within 10-14 days a dog should show symptoms of the disease if infected, so only a two-week holding period is necessary for identifying a potential infection in dogs. Humans have a highly variable incubation period (a few weeks to many months), which depends on the following factors: the depth of the infected wound, the host's immune system, location of infection, and the number of viruses transmitted.

Bloom's Taxonomy: 1-2: Remembering/Understanding

Chapter Section: 31.1

2) Where does the rabies virus reproduce? Using a specific example, explain the implication this has with the virus's incubation time and infection site.

Answer: The rabies virus propagates inside the brain, where once in high numbers can cause disease. This is an important consideration, because if a rabid dog bites an individual on the toe, the virus will likely have a longer incubation time than if the infection site was on his/her shoulder (nearer to the brain).

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 31.1

3) What is the primary cause of viral hemorrhagic fever in the United States? What is the typical cause of death for this disease?

Answer: Hantaviruses usually cause viral hemorrhagic fevers in the United States. Bleeding (or hemorrhaging) is rarely the cause of death for this disease; it is instead usually organ system failure caused by hantavirus that kills an individual. Vascular systems of organs are disrupted by the pathogen that also decreases oxygen and waste transport as well as temperature control.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 31.2

4) During outdoor activities, it is not uncommon to be bitten by mosquitoes several times, so how does an individual know he/she has been infected with the West Nile virus? How would an infected individual be treated for the disease?

Answer: The West Nile virus has an incubation period of 3-14 days, so symptoms should appear during that time. Characteristic symptoms such as West Nile encephalitis or meningitis occur in <1% of infected people. More common symptoms, which could indicate West Nile virus infection, include all of the following: fever, headache, lymphadenopathy, malaise, myalgia, nausea, and rash. No antiviral drugs that exist are effective against West Nile virus, so rest, fluids, and symptomatic relief of fever and pain are treatments for the West Nile virus disease.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 31.6

5) The Black Plague has killed millions of humans over the course of history, yet we do not vaccinate against this disease. Why is this and how is the disease controlled in modern human populations?

Answer: Plague is a zoonosis of rodents that is transmitted between hosts by the rat flea *Xenopsylla cheopis*. Before the host and vector of the plague were known, humans were accidental hosts that were infected in crowded areas where humans and rodents lived close together (such as rat-infested ships, crowded cities, or military encampments). Control of rodent populations has stopped the spread of plague to humans from the natural host/vector of the rat and rat flea. The causative agent of plague, *Yersinia pestis*, is still present in natural populations of rodents and has NOT been eradicated. If a human is infected with *Y. pestis* the infection can be treated with common antibiotics. No vaccine is warranted, because the disease is now very rare in humans and can be treated successfully.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 31.7

6) How is climate change causing the emergence of zoonotic and vector-borne in new human populations?

Answer: Climate change may be changing the range of certain animals and insects that carry diseases that can infect human populations. This can be very serious because when a new disease is introduced into a population that population does not herd immunity, meaning that the disease could potentially spread rapidly. For example, that the increased temperature and rainfall caused by climate change could expand the range of *Aedes aegypti*, the mosquito that carries dengue fever, yellow fever, and malaria.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 30.5

7) What key features of *Bacillus anthracis* makes the anthrax disease impossible to eradicate?

Answer: *Bacillus anthracis* is a soil microorganism that forms spores and can survive for long periods of time under adverse conditions. Spores are very resistant to drying, heat, or cold. It would be impossible to kill the *B. anthracis* cells in soil around the world, so the disease cannot be eradicated. It is controlled by vaccinating livestock animals to protect livestock populations. Some high-risk people who may come into contact with contaminate materials or animals are also vaccinated. Anthrax is a rare disease among the general population, but it can never be completely eradicated.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 31.8

8) Compare and contrast dengue fever and West Nile fever. Which disease would be easier to eradicate and why?

Answer: Both dengue fever and West Nile fever are caused by flaviviruses and are arboviruses because they are viruses spread by a biting insect. Mosquitoes are the vector for both diseases. However, different species spread each disease and these species have different geographic ranges. The two diseases also differ in their reservoir. The reservoir for West Nile virus is wild bird populations, in which mosquitoes transfer the virus between infected birds. Humans are an accidental host and usually do not develop high number of virus particles in their blood, meaning that they are not able to spread the virus. The only known reservoir for the dengue virus is infected humans, so the dengue virus would be easier to eradicate. If vaccination became available and was used sufficiently, the virus would no longer exist in nature. West Nile virus has a very large reservoir in wild animals, making it much more difficult to eradicate.

Bloom's Taxonomy: 5-6: Evaluating/Creating

Chapter Section: 31.6

9) Using a specific example, explain the term "accidental host."

Answer: When an infectious agent invades a cell that is not necessary for its life cycle, it is considered to be infecting an accidental host. For example, *Yersinia pestis* has no life cycle dependency on humans and normally causes disease in rodents. However it is possible for *Y. pestis* to infect a human (the accidental host). Answers could include many other examples from the chapter, such as hantaviruses.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 31.6

10) During a camping and hiking expedition in the Ice Age National Scenic Trail at Hartman Creek State Park (Waupaca, WI), an individual notices a small tick on his arm. After removing the partially embedded deer tick, what symptoms should this individual look for to indicate infection of *Borrelia burgdorferi*? Also, what should be done on future hiking trips to decrease the likelihood of *B. burgdorferi* infection?

Answer: Acute Lyme disease symptoms include backache, the chills, headache, and overall fatigue. Chemicals (e.g., DEET) that repel insects such as deer ticks should be applied all over the individual's body. Deer ticks are also attracted to white from the white-tailed deer they infect, so dark-colored clothing can also minimize *B. borgdorferi* infections. Long protective clothing should also be worn such as socks being tucked into boots and wearing snug collars and cuffs.

Bloom's Taxonomy: 3-4: Applying/Analyzing

Chapter Section: 31.4